

ERDENETSAGAAN, Mongolia

WMO: 443170

Lat: 45.9067N

Lon: 115.3719E

Elev: 1076

StdP: 89.05

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.7	-27.5	-33.0	0.2	-29.4	-31.0	0.2	-27.2	14.2	-16.8	12.3	-17.2	3.7	320	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	31.9	19.0	29.7	17.8	27.7	16.9	20.5	29.2	19.1	27.1	18.0	25.7	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	14.4	24.8	16.3	13.2	22.9	15.1	12.3	21.9	64.6	29.2	59.4	27.1	55.4	25.7	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.8	9.9	-32.6	34.7	2.0	2.6	-34.0	36.6	-35.2	38.1	-36.3	39.6	-37.8	41.4
			-32.4	22.1	1.8	1.9	-33.7	23.4	-34.7	24.5	-35.7	25.5	-37.0	26.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-18.2	-14.9	-6.1	3.7	11.4	17.5	20.6	18.7	12.2	3.0	-8.3	-16.4	(6)
	DBStd	14.72	5.23	5.98	6.62	5.66	5.17	4.20	3.31	3.78	4.80	5.65	6.42	5.22	(7)
	HDD10.0	3946	874	697	501	201	46	1	0	1	31	225	550	819	(8)
	HDD18.3	6142	1133	931	759	438	223	65	13	40	189	474	800	1078	(9)
	CDD10.0	1032	0	0	0	12	88	225	328	271	98	9	0	0	(10)
	CDD18.3	187	0	0	0	0	7	39	83	52	6	0	0	0	(11)
	CDH23.3	1996	0	0	0	4	131	439	785	556	81	1	0	0	(12)
	CDH26.7	649	0	0	0	0	33	149	274	179	13	0	0	0	(13)
	WSAvg	4.2	4.3	4.3	4.6	5.0	5.3	3.7	3.2	3.2	3.9	4.1	4.3	4.4	(14)
(15) Precipitation	PrecAvg	253	2	2	6	8	19	38	68	63	31	8	4	3	(15)
	PrecMax	427	8	8	18	42	51	89	190	142	82	27	10	7	(16)
	PrecMin	114	0	0	0	0	0	6	16	10	2	0	0	0	(17)
	PrecStd	55	2	2	5	10	15	22	39	36	21	8	3	2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.3	4.0	14.6	23.4	30.4	33.9	35.2	34.6	28.6	21.5	11.3	0.5	(19)
		MCWB	-5.2	-0.4	5.6	11.0	15.0	19.0	21.1	20.5	15.9	11.0	4.5	-2.3	(20)
	2%	DB	-5.8	0.3	10.6	19.4	26.8	30.5	32.1	31.0	25.7	18.2	7.1	-4.0	(21)
		MCWB	-8.0	-3.0	3.5	8.9	13.3	17.2	19.8	18.8	14.4	9.7	1.9	-6.1	(22)
	5%	DB	-8.4	-2.8	7.0	16.4	24.1	28.0	29.7	28.4	23.3	15.0	3.7	-6.6	(23)
		MCWB	-9.9	-5.2	1.3	7.4	12.3	16.6	18.4	17.5	13.1	7.5	-0.3	-8.2	(24)
	10%	DB	-10.5	-5.7	3.6	13.6	21.3	25.6	27.5	26.4	20.9	12.3	1.0	-8.9	(25)
		MCWB	-11.7	-7.6	-0.8	5.6	10.9	15.4	17.8	16.4	12.0	5.9	-2.0	-10.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.9	0.0	6.2	11.7	16.0	20.7	23.2	21.8	17.4	12.4	4.6	-2.6	(27)
		MCDB	-2.4	3.2	14.0	21.4	28.3	31.5	33.0	31.1	26.2	19.8	10.8	0.6	(28)
	2%	WB	-7.8	-3.0	3.6	9.5	14.2	18.7	21.0	20.1	15.3	9.7	2.1	-6.1	(29)
		MCDB	-5.9	0.2	10.2	18.4	25.3	28.0	29.3	29.3	23.6	17.6	7.0	-4.2	(30)
	5%	WB	-9.9	-5.2	1.5	7.6	12.9	17.4	19.6	18.5	13.9	7.7	-0.2	-8.2	(31)
		MCDB	-8.5	-2.7	6.7	15.8	22.6	25.8	27.0	26.4	21.6	14.3	3.6	-6.6	(32)
	10%	WB	-11.7	-7.6	-0.7	5.9	11.5	16.2	18.5	17.2	12.6	5.9	-2.0	-10.2	(33)
		MCDB	-10.6	-5.8	3.4	13.6	20.3	23.8	25.7	24.4	19.7	11.9	1.0	-8.9	(34)
(35) Mean Daily Temperature Range	MDBR		8.9	10.3	11.2	12.4	12.9	11.4	10.9	11.9	12.3	11.6	10.0	8.4	(35)
	5% DB	MCDBR	11.9	12.9	14.3	16.0	16.6	14.4	13.7	14.2	14.8	14.7	12.6	10.9	(36)
		MCWBR	10.4	10.4	9.3	9.6	8.8	6.9	6.4	6.8	8.0	9.4	9.1	9.4	(37)
	5% WB	MCDBR	11.8	12.6	13.9	15.1	15.4	12.7	12.1	12.4	13.5	14.2	12.6	10.7	(38)
		MCWBR	10.4	10.3	9.5	9.5	8.6	6.8	6.5	6.7	7.8	9.5	9.2	9.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.254	0.278	0.324	0.379	0.447	0.462	0.416	0.387	0.344	0.313	0.276	0.259	(40)
	taud		2.328	2.285	2.216	2.114	1.960	2.038	2.246	2.293	2.352	2.359	2.341	2.298	(41)
	Ebn at Noon		861	902	900	874	823	811	847	859	871	848	820	806	(42)
	Edh at Noon		74	96	120	147	178	165	133	122	105	89	74	69	(43)
(44) All-Sky Solar Radiation	RadAvg		2.14	3.29	4.70	5.80	6.49	6.35	6.16	5.75	4.72	3.38	2.22	1.72	(44)
	RadStd		0.14	0.21	0.18	0.29	0.40	0.47	0.28	0.29	0.28	0.20	0.13	0.11	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	-1.74	N/A	-0.58	-0.86	N/A	N/A	N/A	N/A

Nomenclature: See separate page